THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Life Processes in Plants** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A leaf is kept in the dark for 48 hours, then half of it is covered with black paper and kept in sunlight for 6 hours. Which statement is correct? 1

(a) the covered half remains brown with iodine
(b) the covered half turns blue-black
(c) both turn blue-black
(d) both remain brown

2. Which of these energy changes occurs in photosynthesis? 1

(a) heat to light **(b)** sound to light
(c) light energy to chemical energy **(d)** chemical to light

3. A farmer notices that his legume plants in poor soil have swollen nodules in roots. This indicates: 1

(a) water logging **(b)** nitrogen-fixing bacteria
(c) insects **(d)** fungi attack

4. A variegated leaf kept in sunlight: only the green part turns blue-black. This shows: 1

(a) CO2 is not needed **(b)** chlorophyll is needed
(c) water is not needed **(d)** light is not needed

5. Which energy change occurs in respiration? 1

(a) sound to chemical **(b)** chemical energy to usable energy
(c) light to chemical **(d)** heat to chemical

6. A farmer grows wheat after pulses and applies less urea. The reason is: 1

(a) wheat needs no nitrogen **(b)** urea is costly
(c) nitrogen left by pulses **(d)** pulses consume nitrogen

7. In a set-up with a plant in a bell jar with NaOH solution, starch is absent in the leaf because: 1

(a) light is blocked **(b)** NaOH gives CO2
(c) NaOH adds oxygen **(d)** NaOH absorbs carbon dioxide

8. Which of these is a net result of photosynthesis in a day? 1

(a) no glucose **(b)** only oxygen
(c) less glucose **(d)** more glucose than used in respiration

9. A student claims that mushrooms make food in sunlight. The correct reply is: 1

(a) they make starch
(b) correct
(c) mushrooms have no chlorophyll and absorb food from dead matter
(d) they use CO2

10. In an experiment, aquatic plant produces 10 bubbles per minute at 30 cm from a lamp and 40 at 10 cm. This shows photosynthesis rate: 1

(a) decreases with light **(b)** increases with light intensity
(c) stops at night **(d)** is independent of light

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a plant in a closed jar with a burning candle and plant show that plants restore the air? 2

12. Explain why the roots of water-logged plants rot. 2

13. How do plants adapt to deserts to reduce transpiration? 2

14. Explain why the leaf in the dark does not show starch though it is green. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Design an experiment to show that oxygen is released during photosynthesis, with controls. 3

16. Why do trees in a dense forest have most of their leaves at the top? 3

17. Why does an aquatic plant show more bubbles on a bright day? 3

OR

Explain the changes in a leaf's starch content from morning to night and the reason. 3

18. Explain with examples how plants help in keeping the air clean. 3

OR

Explain how leaf shape and size differ in different habitats and why. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design a fair test to find the effect of light intensity on the rate of photosynthesis using hydrilla and a lamp. Name the variables, steps and expected graph. 5

OR

Describe an investigation using three potted plants (light/dark, with/without water, with/without CO2) to study the factors of photosynthesis. 5

20. A plant biologist finds a plant with no chlorophyll which grows on the bark of a tree. Discuss the possible modes of nutrition and how to decide. 5

OR

Compare the roles of respiration and photosynthesis in the carbon-oxygen cycle. 5

END OF QUESTION PAPER — Total: 40 Marks